



## LAB 3

# LAB 3 — MACHINE LEARNING ON DATA REPRESENTATIONS

How much performance comes from the model —  
and how much from the representation?

PRESS – OR SPACE. →

# The Task

- Classify land cover from geospatial data assembled in Lab 2
- Work with point-based samples from raster/geospatial layers
- Use one dataset consistently across all experiments
- Evaluate predictions on held-out data

Implementation is yours: Python in **Google Colab**, **VS Code**, or local Jupyter. CPU is enough.

# Experimental Questions

Q1

Does feature engineering help?

Q2

Does complementary information help?

Q3

When do we need a nonlinear model?

Treat this as an experiment: hold conditions fixed and change one conceptual factor at a time.

# Build the Dataset Once

- Choose/extract labeled sample locations
- Build one table: **rows = locations**, **columns = features**
- Target column: land-cover class
- Create one fixed train / validation / test split

**Important:** Do the split once and reuse it for every experiment.

# Baseline: Logistic Regression

- Use `LogisticRegression` as the linear baseline
- Input: feature vector per sample
- Learns a linear decision boundary
- Fast, simple, and interpretable reference model

**Key question: How good is your representation if linear separation is all you allow?**

# Change the Representation (Model Fixed)

Retrain the same `LogisticRegression` while changing features:

1. Small raw feature set (e.g., Red + NIR)
2. Raw features + engineered feature (e.g., NDVI)
3. Several Sentinel-2 bands
4. Bands + complementary Lab 2 variables (if available)

Prompt: What changes when the model stays fixed but the representation changes?

# Compare with Random Forest

- Run `RandomForestClassifier` on the same representations
- Keep the exact same train/validation/test split
- Compare against logistic regression fairly

## Logistic Regression

Linear decision boundary

## Random Forest

Nonlinear decision boundaries

# Ablation Table

| Representation           | Logistic Regression | Random Forest |
|--------------------------|---------------------|---------------|
| Raw baseline             | ...                 | ...           |
| + NDVI                   | ...                 | ...           |
| + additional bands       | ...                 | ...           |
| + complementary features | ...                 | ...           |

Use accuracy (and optionally macro-F1). Focus on interpreting changes, not maximizing one score.

# Interpret the Experiment

- Which representation change helped most?
- When did logistic regression approach random-forest performance?
- What does the model gap suggest about feature-space geometry?

**Caution:** More dimensions are not automatically better. Useful, complementary information matters.

# Takeaway → Lecture 4

1. Feature engineering changes the representation.
2. Complementary information can make classification easier.
3. A strong representation can make simple linear models surprisingly powerful.

**NEXT**

**In Lab 3 you designed representations manually.  
In Lecture 4: what if the model learns the representation itself?**